Questions

Q1.

A class is learning about refraction of waves.
The teacher shows them a demonstration using a battery-powered toy car travelling across a smooth road and onto some sand.
The car slows down as it enters the sand.

Figure 7 shows the car just before it meets the sand.

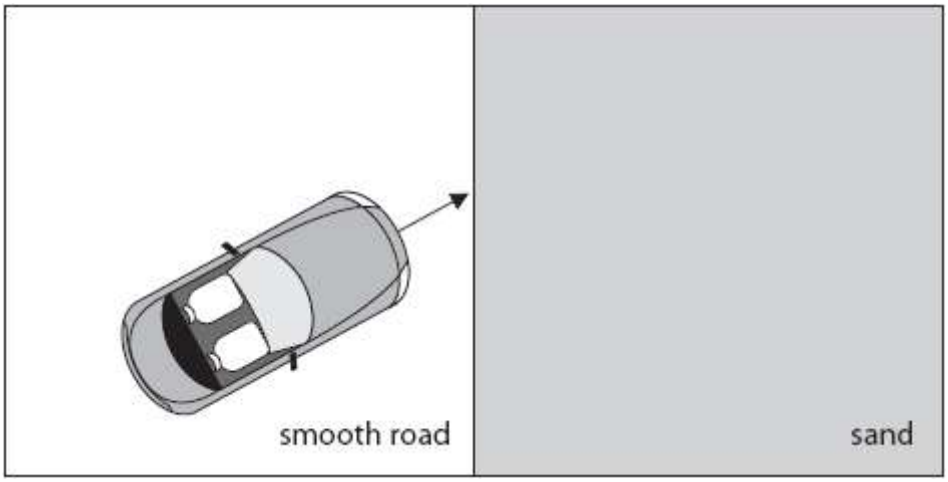


Figure 7

(i) Draw an arrow on the diagram to show the direction of the car as it travels across the sand.

(1)

(ii) Explain why this is a useful model for refraction of light.

(2)

.....

.....

.....

.....

(Total for question = 3 marks)

Q2.

Sound travels slower in air than it does in water.

Figure 9 shows the direction of travel of a sound wave approaching a boundary between air and water.

The sound wave refracts at the boundary between air and water.

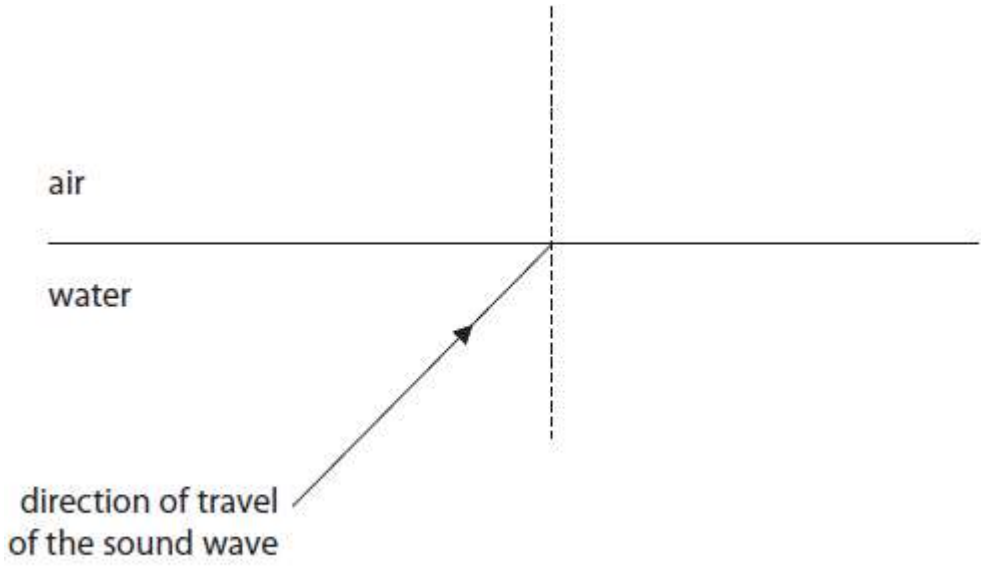


Figure 9

Complete the diagram in Figure 9 to show the direction the sound wave travels in the air.

(2)

(Total for question = 2 marks)

Q3.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The human ear can only detect frequencies below ultrasound and above infrasound.

(i) Which of these gives the approximate range of frequencies for the human ear?

(1)

☐ **A** 2 kHz to 2 Hz
☐ **B** 20 kHz to 20 Hz
☐ **C** 2 000 kHz to 2 kHz
☐ **D** 200 kHz to 20 kHz

(ii) Suggest **two** reasons why there are limits to the frequencies that a human ear can detect.

(2)

1 .....

.....

2 .....

.....

(Total for question = 3 marks)

Q4.

This question is about different types of electromagnetic radiation.

The human eye can detect a range of wavelengths, from violet at about 400 nm to red at about 700 nm. Figure 8 shows the relative sensitivity of the human eye to different wavelengths when inside a dark cave.

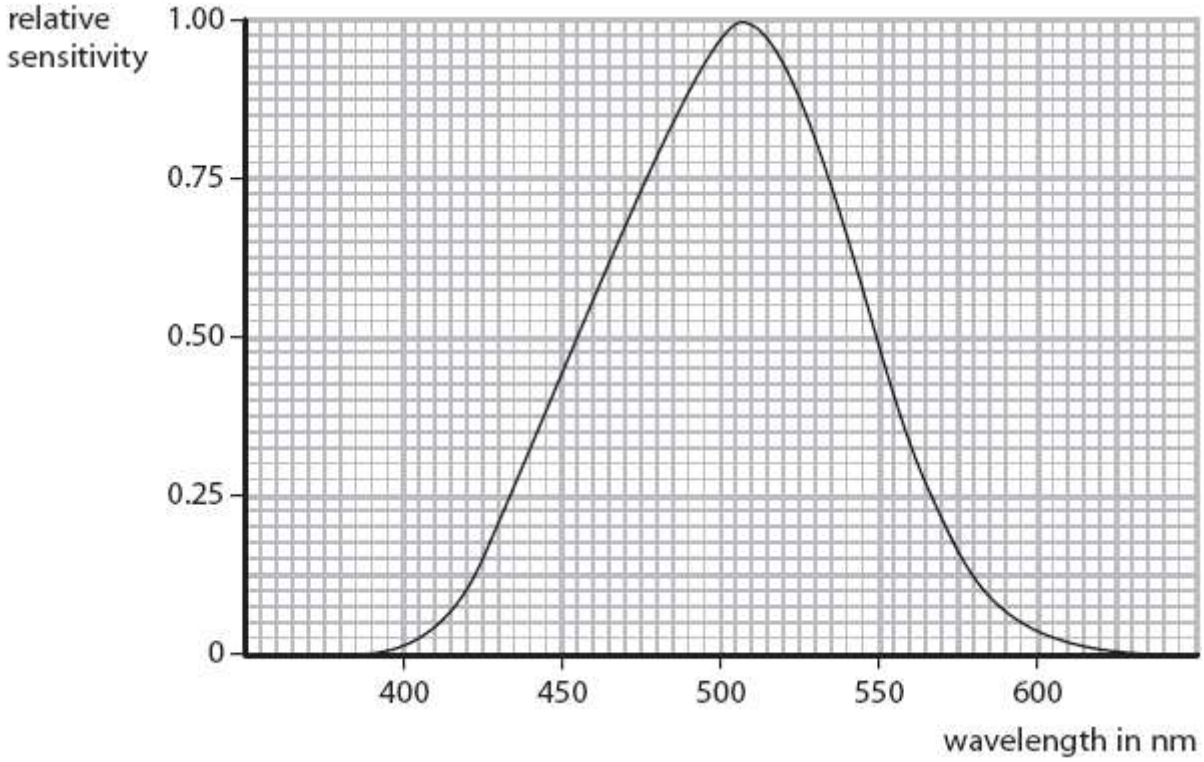


Figure 8

(i) Use the information in Figure 8 to determine the wavelength to which the eye is most sensitive inside the cave.

(1)

wavelength = ... nm

(ii) Calculate the frequency to which this wavelength corresponds. Give the unit.

The speed of light in air = 3.0×10^8 m/s.

(3)

frequency = ... unit .....

(iii) Using information from Figure 8, describe how well the eye detects different colours of the visible spectrum inside the cave.

(3)

.....

.....

.....

.....

.....

.....

(Total for question = 7 marks)

Q5.

A sound wave travels with a velocity of 1530 m/s.

The frequency of the wave is 1.20 kHz.

Calculate the wavelength of the wave.

(3)

wavelength = ... m

(Total for question = 3 marks)

Q6.

This question is about waves in the electromagnetic (e.m.) spectrum.

(i) A microwave oven uses waves of frequency 2.45 GHz.

Calculate the wavelength of the microwaves.
The velocity of light is 3.00×10^8 m/s.

(3)

wavelength = ... m

(ii) The microwave oven is 55% efficient and transfers 42 000 J of energy to some food when it is heated. Calculate the total amount of energy that must be supplied to the oven.

(3)

energy supplied to oven = ... J

(Total for question = 6 marks)

Q7.

Figure 1 shows part of a wave.

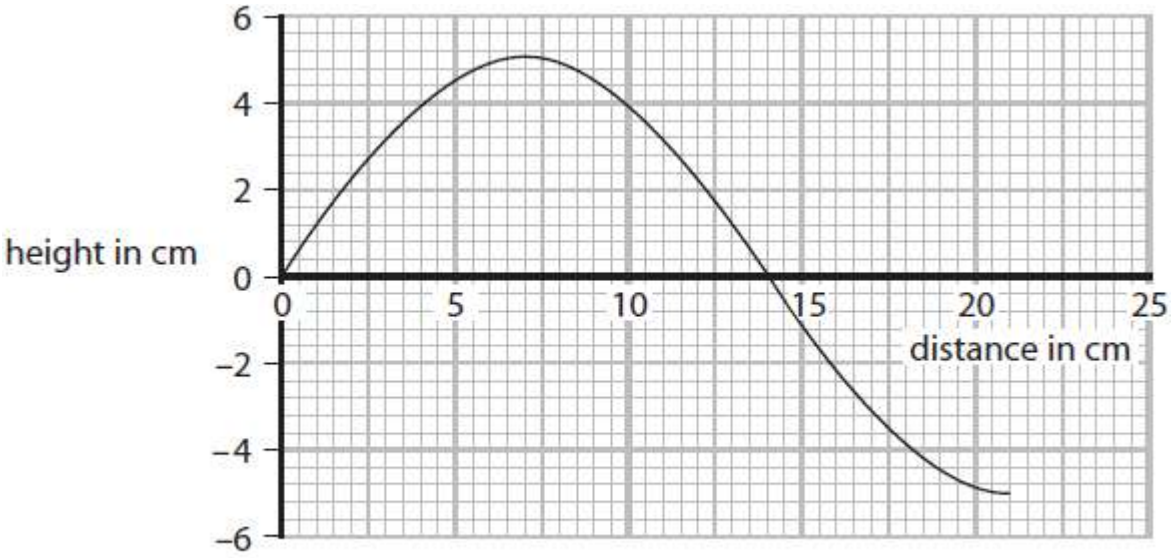


Figure 1

Use data from Figure 1 to calculate the wavelength of the wave.

(2)

wavelength = ... cm

(Total for question = 2 marks)

Q8.

A different water wave has a wavelength of 0.25 m and a frequency of 1.5 Hz.

Calculate the wave speed.

(2)

wave speed = ... m/s

(Total for question = 2 marks)

Q9.

This question is about both **infrasound** and **ultrasound**.

Figure 1 shows a technician standing in front of a wall.

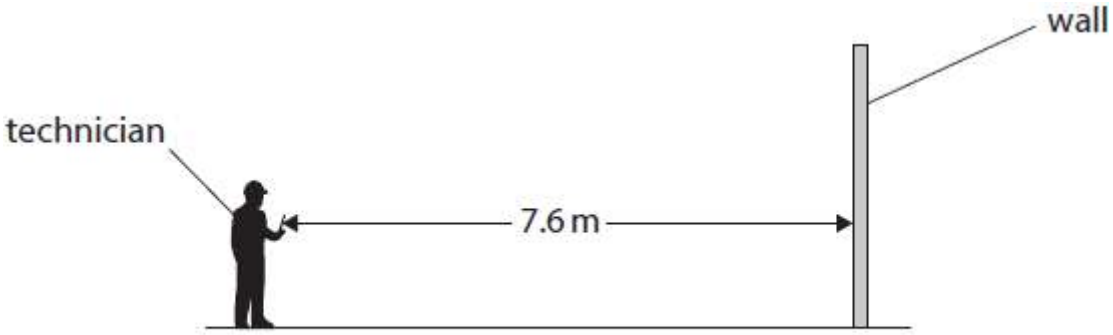


Figure 1

The technician has a device that can emit and detect **ultrasound**.

The device emits a pulse of **ultrasound** towards the wall and detects the echo of the pulse a short time later.

The wall is 7.6 m from the device.

The speed of **ultrasound** in air is 330 m/s.

Calculate the time between the emission of the pulse and the detection of the echo.

(3)

time = ... s

(Total for question = 3 marks)

Q10.

A radio station transmits on 97.4 MHz.

To receive the waves an aerial needs a length equal to half the wavelength of the radio waves being transmitted.

Calculate the length of the aerial needed.

The speed of the radio waves is 3.00×10^8 m / s.

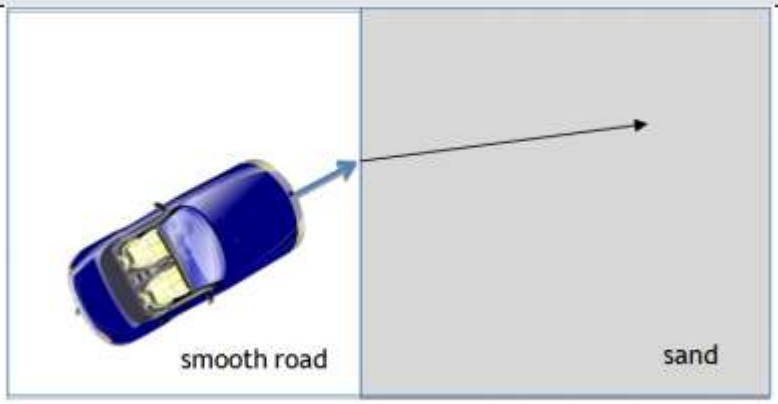
(3)

length of aerial = ... m

(Total for question = 3 marks)

Mark Scheme

Q1.

Question number	Answer	Mark
(i)	 approx	(1)

Question number	Answer	Additional guidance	Mark
(ii)	An explanation that combines identification - knowledge (1 mark) and reasoning/justification - understanding (1 mark): • both car and light ray slow down when entering sand / glass (1) • direction changes towards normal (1)	Bend towards the normal	(2)

Q2.

Question Number	Answer	Additional guidance	Mark
	single straight line in upper right quadrant (1) direction change **towards** the normal (1)	ignore arrow direction conditional on first mark point	(2)

Q3.

Question Number	Answer	Mark
	B 20k Hz to 20 Hz ***A*** *is not correct because 2Hz is an infrasound frequency* ***C*** *is not correct because 2000kHz is an ultrasound frequency* ***D*** *is not correct because 200kHz is an ultrasound frequency*	**(1)** **AO1**

	Answer	**Additional guidance**	**Mark**
(ii)	An explanation linking any two of :-		**(2)** **AO2**
	the size of the parts of the (inner) ear (1)	allow length of cochlea	
	vibrations at (limited) range of frequencies/range of sounds most common as adapted through evolution (1)		
	(membrane) variations in stiffness/ width/thickness (1)	allow cochlea	
	different parts vibrate at different frequencies (1)		
	eardrum not sensitive enough to detect low/high frequencies (1)	allow ultrasound / infrasound	
	brain cannot interpret low/high frequencies (1)	allow ultrasound / infrasound	

Q4.

Question number	Answer	Additional guidance	Marks
(i)	Accept any answer between 502 and 510 nm inclusive		1

Question number	Answer	Additional guidance	Marks
(ii)	Substitution (1) frequency = $3.0 \times 10^8 / (505 \times 10^{-9})$ Evaluation (1) $=5.9 \times 10^{14}$ Unit mark (1) Hz	 accept answers in range 5.88×10^{14} to 5.98×10^{14} Accept hertz / s^{-1}	(3)

Question number	Answer	Additional guidance	Marks
(iii)	An answer that combines three points of interpretation/evaluation to provide a logical description (3) Yellow or green or blue detected/seen the best/most sensitive to / highest relative sensitivity (1) Red or orange not detected/not seen (so well)/least sensitive to/lowest relative sensitivity (1) violet or indigo not detected/not seen (so well)/less sensitive to (1)	 about 500 nm more than 550 nm less than 450 nm 2 max if only wavelength instead of colours	(3)

Q5.

Question number	Answer	Acceptable	Mark
	substitution (1) 1530 = (1.2/1000) x wavelength rearrangement (1) wavelength = 1530 / (1.2/1000) evaluation (1) = 1.275 (m)	2 marks for 1275m (incorrect conversion into Hz) full marks will be awarded for correct numerical answer without working	**(3)**

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	selection and substitution (1) $3(.00) \times 10^8 = 2.45 (\times 10^9) \times \lambda$ rearrangement (1) $(\lambda=) \frac{3(.00)\times10^8}{2.45 (\times10^9)}$ evaluation (1) 0.12 (m)	allow substitution and rearrangement in either order $2.45 \left(\times10^9\right)=\frac{3(.00)\times10^8}{\lambda}$ $\lambda= \frac{v}{f}$ accept 0.122(m) power of ten error gains 2 marks award full marks for the correct answer without working	**(3)** **AO2**

Question Number	Answer	Additional guidance	Mark
(ii)	selection and substitution (1) $(0.)55=\frac{42\ 000}{\text{total energy supplied (to device)}}$ rearrangement (1) $(\text{total energy supplied to device}=)\frac{42\ 000}{(0.)55}$ evaluation (1) 76 000(J)	allow substitution and rearrangement in either order $(0.)55=\frac{42\ 000}{x}$ accept any value that rounds to 76 000(J) 760/764/763(J) gains 2 marks any other power of ten error gains 1 mark award full marks for the correct answer without working	**(3)** **AO2**

Q7.

Question number	Answer	Additional guidance	Mark
CS1	uses data taken from x axis (1) 28(cm) (1)	 award full marks for correct answer without working	(2) AO3

Q8.

Question Number	Answer	Additional guidance	Mark
	recall and substitution (1) (v =) 0.25 x 1.5 evaluation (1) 0.38 (m/s)	 accept 0.375 or 0.37 (m/s) accept 37.5, 37 or 38 for 1 mark only award full marks for the correct answer without working	(2)

Q9.

Question Number	Answer	Additional guidance	Mark
	selection of speed = $\frac{\text{distance}}{\text{time}}$ (1) rearrangement and substitution (1) time = $\frac{7.6}{330}$ (× 2) evaluation (1) 0.046 (s)	time = $\frac{\text{distance}}{\text{speed}}$ allow values that round to 0.046 e.g. 0.04606 award full marks for the correct answer without working 0.023(03) scores 2 marks	**(3)** **AO2.1**

Q10.

Question number	Answer	Additional guidance	Mark
	recall **and** rearrangement (1) $\lambda = \frac{v}{f}$ evaluation (1) 3.08 (m) (so) length of aerial = 1.54 m (1) check working $\frac{3\times10^8}{2} = 1.5 \times 10^8$ gets only 1 mark for ecf	 $\frac{3.0\ (\times 10^8)}{97.4\ (\times 10^6)}$ accept 3.1 (m) award 1 mark for wavelength that rounds to 3.1 to any other power of 10 independent mark. allow ECF from candidate's wavelength accept 1.5 (m) award 2 marks for 1.5 to any other power of 10 award full marks for the correct answer without working Allow 1.46 rounded to 1.5 for 1 mark only if it is ecf from mp2	**(3)**